

Algorithms Notes - Search

Linear search examines items sequentially and takes $O(n)$ time in the worst case.

Binary search requires sorted input and halves the remaining interval each step.

Sorting

Merge sort divides the input, sorts each half, and merges the results.

Its worst-case running time is $O(n \log n)$ and it needs $O(n)$ auxiliary space.

Graphs

Breadth-first search uses a queue and finds shortest paths in unweighted graphs.

Depth-first search uses a stack or recursion and is useful for cycle detection.

Dynamic Programming

Dynamic programming stores solutions to overlapping subproblems.

Memoization is top-down while tabulation is bottom-up.

Hashing

A hash table provides expected $O(1)$ lookup with a suitable hash function. Separate chaining stores colliding keys in per-bucket collections.